Status	Finished
Started	Wednesday, 28 October 2026, 11:21 PM
Completed	Wednesday, 28 October 2026, 11:35 PM
Duration	13 mins 36 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Complexity

c. Selection

d. Bubble ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists

b. It requires the list to be sorted ✓

c. It does not work with negative numbers

d. It is slower than linear search for small lists

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓

b. By starting from the first element

c. By using a hash function

d. By comparing each element sequentially

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Checking each element sequentially ✓
c. Dividing the list in half repeatedly
d. Skipping every second element

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
d. There must be a mechanism to access middle element directly

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94
b. 89 and 99
c. 89 and 94
d. 90 and 99 ✓

Question **7**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search ends successfully ✓
c. The list is sorted
d. The search continues in the right sublist

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the entire list
c. In the left sublist ✓
d. At the beginning of the list

Question **9**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It can be applied only if the list is sorted ✓
c. It always starts from the beginning of the list
d. It compares elements sequentially

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. The list must be sorted
c. There must be a mechanism to delete and/or insert elements in the list ✓
d. Number values should only be present

Number values should only be present

Question **11**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Finding
c. Discovering
d. Searching ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. 1
c. n ✓
d. log n

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search requires the list to be sorted.
d. Linear search is more efficient than binary search on large lists.

Question **14**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the middle of the list
c. In the left sublist
d. In the right sublist ✓

Question **15**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓
b. Item is the last element in the array or item is not there at all
c. When the item is the last element in the array
d. When the item is not the array at all